EPFL - Computer Science

CS-401 Applied Data Analysis

Lecture 06 — October 15, 2025

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Causal analysis of observational data

Goal of this lecture

- Clarify difference between **experimental** and **observational** studies
- Highlight **pitfalls** of observational studies
- Give you **tools** for avoiding the pitfalls, allowing you to draw valid conclusions from "found data" (very useful for project!)
- Motivate you to **read** Rosenbaum's great book http://www.stewartschultz.com/statistics/books/Design
- in particular Chapters 1, 2, and 3;

Randomized controlled experiments

- Two experimental conditions:
  - Treatment (e.g., medicine)
  - Control (e.g., placebo)
- Assignment of participants to conditions is random
  - Probability of receiving treatment same for everyone
- Treatment and control groups are indistinguishable
  - E.g., determination to quit smoking is not systematically higher in the treated group

Limits of randomization

- Do seat belts save lives?
- Experiment:
  - Flip coin at birth to assign to treatment (always wear set belt for entire life) or control (never wear seat belt)
  - Measure fraction of traffic deaths in each group
- Randomized experiments aren't always feasible

  - Unethical, expensive, fundamentally impossible (e.g., do earthquakes decrease life spans?)
  - Most modern "big data" is "found data"
- Experiments may lead to unrealistic scenarios

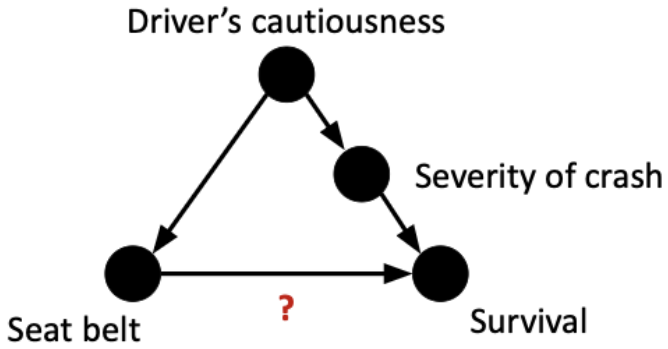
Alternative: observational studies

- Fundamentally different from experiment:
  - Researcher can't control who goes to which condition
  - Researcher is merely an observer, not a tinkerer
  - Much less problematic w.r.t. ethics, price, feasibility
  - Much more problematic w.r.t. validity of conclusions
- All advantages of randomized experiment are gone
  - Subjects self-select to be treated
  - Treatment assignment and response may be caused by same hidden correlate (a.k.a cofounders; e.g., motivation to quit smoking)

Example: seat belts revisited

- Recall: experiment infeasible because unethical
- Observational study:
  - Dataset: all traffic accidents in a given time span
  - Two treatment conditions:
    * Treated: seat-belt wearers
    * Control: non-seat-belt wearers
  - Compare fraction dead in treated vs. control
- What problems do you see?
  Maybe some people are more cautious driver (maybe because they are not using seat belt). Another, the position in the car may influence the likelihood of dying. Age could also be a factor, the car (old cars may not have airbags), etc...

As a causal diagram



A matched observational study

- Consider only a particular subset of accident cars:
  - 2 people in car: driver + passenger
  - Exactly one of them died in accident
  - Exactly one of them wore seat belt at time of accident (i.e., 1 treated + 1 control per car)
- As before: compare fraction dead in treated vs. control
- New: many potential co-founders are controlled for, including type of car, speed, severity of accident
- Fundamental concept: **matching**

Setlling the seat-belt question

	Driver Passenger	Not Belted Belted	Belted Not Belted
Driver Died	Passenger Survived	189	153
Driver Survived	Passenger Died	111	363

Natural experiments

- Not researches, but nature, "flips a coin" to decide treatment assignments
- Examples: twin studies
- Rosenbaum: "When investigators are especially proud, having found unusual circumstances in which treatment assignment, though not random, seems unusually haphazard, they may speak of a 'natural experiment."
- Is matched seat-belt study a natural experiment?

Nature didn't flip a coin for me – should I go home and weep?

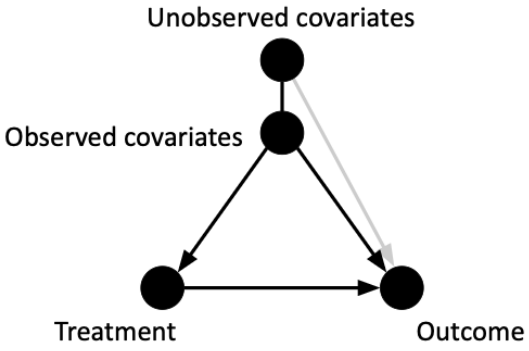
- No! You can still get good mileage if you're smart about it
- Fundamental concept: **matching**
- Ideally: Pair up 2 identical people:
  - 1 treated, 1 control
  - Ex-post (vs. natural experiment: ex-ante)
- Compare outcome of treated vs. control
  - e.g, mean difference treated-minus-control
  - or regression analysis
- Such ideal matching usually not feasible
  - Problem 1: **Unobserved covariates**:
    - You usually can't even know if two people are identical
    - e.g., (hypothetical) gene that causes both desire to smoke and lung cancer
  - Problem 2: **Combinatorial explosion**:
    (Nearly) no two people are identical

Addressing Problem 1 by ignoring it: A naive model

"People who look comparable are comparable"

or equivalently:

"Only observed covariates determine treatment assignment"



Unobserved covariates can't affect the treatment! Furthermore, we don't care how observed and unobserved covariates are related.

Example:

- Unobserved covariates: motivation to quit smoking

- Observed covariates: gender of people

Note: Some gender may have a higher motivation to quit smoking, but this doesn't affect the treatment directly!

If the naive model was true...

- ... you could "simulate" a randomized experiment:
  - Simply match subjects with identical observed covariates (1 treated, 1 control)
  - Subjects in a pair have the same probability to treat
  - So who gets treated is up to chance, as in experiment
  - Analysis: compare outcome for treated to outcome of control (e.g., mean difference treated-minus-control)

Problem 2: Combinatorial explosion

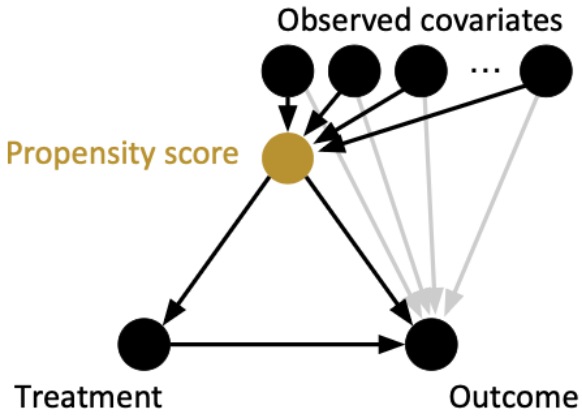
- (Nearly) no two people are identical
- So finding two people to match is often impossible
  - Even when considering only observed covariates (as in the naive model)
- Do we really need to match people with identical **covariates**?
  - Recall "holy grail": randomized controlled trials
  - Coin makes sure everyone has identical **probability to be treated**
  - $\Rightarrow$ Let's mimic what the coin does!

Addressing Problem 2: Propensity score

- Compress the (potentially many) observed covariates into a single number: the probability to receive the treatment (a.k.a. propensity score):

$$\texttt{Pr(subject is treated} \mid \texttt{observed covariates)}$$

- Can be estimated from the data
  - E.g., via logistic regression
  - Input: observed covariates
  - Output: treatment indicator (1 if treated, 0 if control)



Balancing property of propensity score

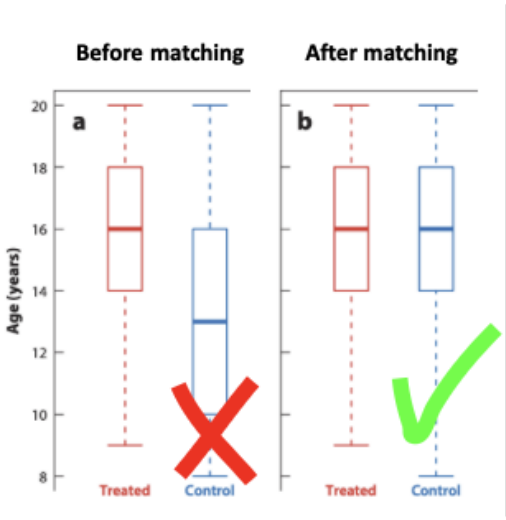
- Fact: all subjects (treated and control) with equal propensity score (PS) p have equal distribution of observed covariates $\mathbf{x}$:

$$\Pr(\mathbf{x} \mid \text{treated} = 1, \text{PS} = p) = \Pr(\mathbf{x} \mid \text{treated} = 0, \text{PS} = p)$$

- Subjects in a matched pair might not have equal $\mathbf{x}$, but treated and control groups will have similar distributions of $\mathbf{x}$

Balancing property is propensity score's reason to be

- There are many other methods for achieving balance
  - e.g., exact matching, Mahalanobis distance matchings, coarsened exact matching...
- You can mix and match methods (e.g., match exactly on gender, use propensity score for other covariates)
- What eventually matters is whether you achieve balance
  - Regardless of *how you try to achieve balance*, you need to verify *that you managed to achieve balance*

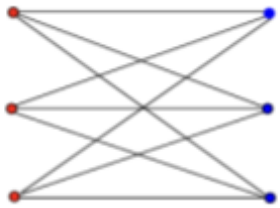


	Treated	Control	Unmatched
Female %	67.3	67.3	46.3
Age (mean)	14.9	14.7	13.5
Black %	14.3	15.1	28.1
Hispanic %	14.3	14.9	39.9
BMI (mean)	22.3	22.4	22.5
BMI missing %	4.1	2.2	0.0
Family income/poverty (mean)	2.4	2.4	2.1
Family income/poverty missing %	0.0	0.6	7.0
$\log_{10}$ (1 + cotinine)	0.4	0.4	0.3
Cotinine missing	2.0	2.7	11.1
Propensity score (mean, as a %)	1.7	1.7	0.7

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Matching algorithms

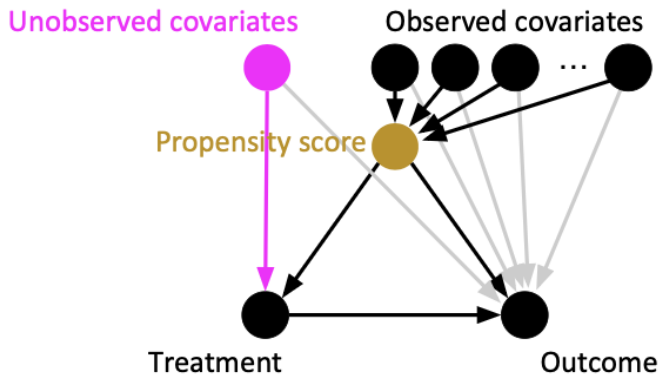
- Goal: Match subjects into pair (1 treated, 1 control), with identical propensity scores within each pair
- Unlikely that 2 subjects have identical propensity scores
- Use approximate matching (remember your algo class!)
- Bipartite graph: each subject connected to all other subjects
- Edge weights: absolute (or squared) difference of propensity scores (or other matching criterion)
- Find minimum matching, e.g., via Hungarian algorithm



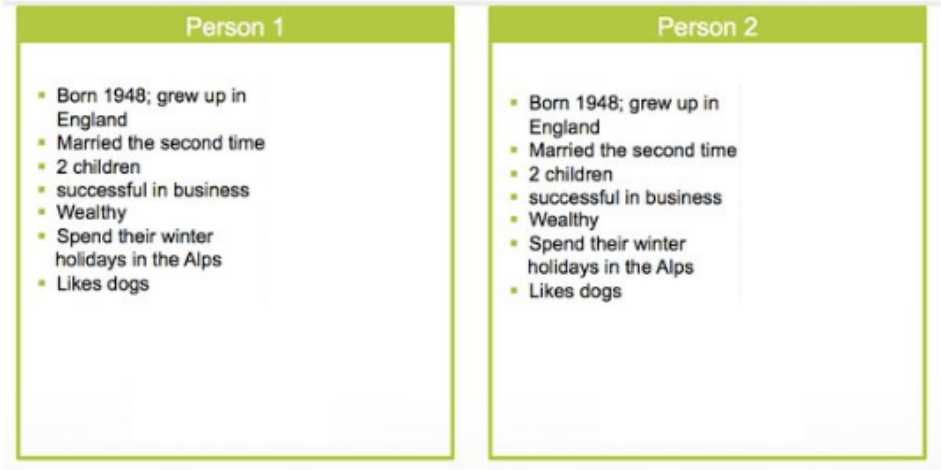
If the naive model isn't true...

- ... propensity score may differ from true probability to treat:

$$\Pr(\text{treated} \mid \text{observed covariates}) \neq \Pr(\text{treated} \mid \text{all covariates})$$



If the naive model isn't true, you may end up matching 2 peoples that are different but have similar description...



The sensitivity analysis model

- **Idea**: Quantify the degree to which the naive model may be wrong without you having to change your (casual) conclusions
- Assume that treatment odds of identical-looking-subjects (i.e., identical observed covariates $\mathbf{x}$) may differ by up to a factor Γ
- Then reason in spirit of proof by contradiction: "To change the conclusions of my study, two identical-looking people (1 treated, 1 control) would have to have hugely different treatment odds (i.e., huge Γ). Common sense (or domain knowledge) suggests that this is no the case, so my conclusions stand".

$$\frac{1}{\Gamma} \leq \frac{\pi_k/(1-\pi_k)}{\pi_l/(1-\pi_l)} \leq \Gamma \text{ whenever } \mathbf{x}_k = \mathbf{x}_l.\ \Gamma \geq 1.$$

- Bounded odds ratio (OR)

  - Reason for using OR:
    OR = Pr(k treated | **either k or l treated**) / Pr(l treated | **either k or l treated**)

- Sensitivity $\Gamma = 1 \Rightarrow$ naive model is true
- Sensitivity $\Gamma = 2 \Rightarrow$ subject with same observed covariates $\mathbf{x}$ up to twice as likely to be the one to receive treatment
- Sensitivity $\Gamma = \infty \Rightarrow$ avoid statement (a.k.a. tautology)

Example: smoking and lung cancer

- Under naive model: matching on observed covariates gives a very small p-value for the null hypothesis that smoking does not increase lung cancer risk (using appropriate hypothesis test), i.e., data hard to explain without a causal effect
- Tobacco lobby: "The naive model isn't true! There may be a hidden (e.g., genetic) correlates that increase both the probability to enjoy smoking and the probability of lung cancer. They, not smoking, cause cancer!
- Under sensitivity analysis model, increasing sensitivity Γ increase the p-value for null hypothesis
- Anti-tobacco lobby: "But making $p > 0.05$ would require $\Gamma > 6$; i.e., the odds of being a smoker would need to be six times higher for one of two people with the exact same observed features (age, gender, education, income, ...). It's unlikely that any unobserved covariate would have such a large effect on smoking habits. So smoking causes cancer!"

Two parts: mechanical vs. scientific

- Mechanical part:
  - Create pairs (1 treated + 1 control) with similar observed covariates (using exact or propensity-score matching)
- Scientific (i.e., fun) part:
  - Mitigate concerns that your findings might be caused by unobserved covariates, rather than treatment (e.g., using sensitivity analysis, ad-hoc arguments, natural experiments)

Summary

- Holy grail: randomized experiment
- When experiment not possible: observational study
- Crucial problem: treatment assignment not random (biases!)
- Semi-holy grail: natural experiment
- Matches studies: pair up treated/control based on observed covariates
- Problem: still, treatment assignment not random (biases via unobserved covariates)
- Solution: sensitivity analysis
- Keep this lecture in mind for your projects!